Roll No. ....

TEE–502

B. TECH. (ELECTRICAL ENGINEERING)
(FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

CONTROL SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Summarize the limitations of open loop systems over closed loop systems and advantages of closed loop systems over open loop systems.

10 Marks (CO2)

OR

(b) Explain (in brief) the following terms as used in control systems :

(i) Final Control Element

(ii) Control Variable and Controlled Variable

10 Marks (CO2)

2. (a) What do you mean by transfer function ? What are the various methods to obtain the transfer function of any given system ? 10 Marks (CO1)

OR

(b) Apply the Laplace transform on the following differential equation :

$$\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 5y = 1$$

10 Marks (CO1)

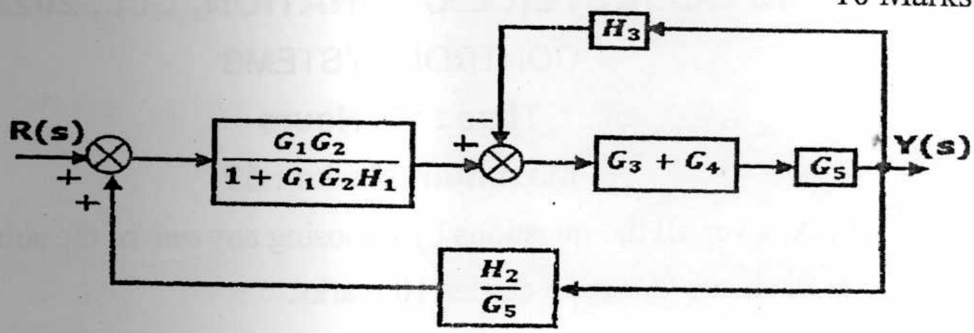
P. T. O.

3. (a) Develop the transfer function of an electric circuit in which a resistance (R), a capacitor (C) and an Inductor (L) are connected in series across an ac supply $v(t) = V_m \sin(\omega t)$. 10 Marks (CO3)

OR

(b) Apply the block diagram reduction rules and obtain the overall transfer function of the system represented by the following block diagram :

10 Marks (CO3)



4. (a) Analyze the time response of a first order system, as expressed by the equation below, for the unit ramp input : 10 Marks (CO4)

$$C(s) = \frac{1}{0.5s + 2} R(s)$$

where, $R(s) = \frac{1}{s^2}$.

OR

(b) The open loop transfer function of a unity feedback system is

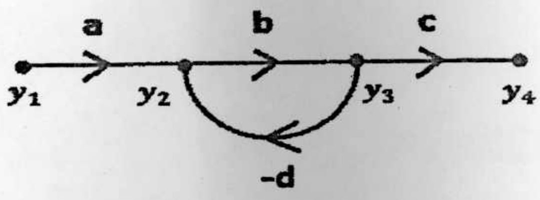
$$G(s) = \frac{4}{s(s+1)}$$

$$H(s) = 1$$

Determine the nature of response of the closed loop system for a unit step input. Also, determine the rise time of the given system.

10 Marks (CO4)

5. (a) Obtain the overall transmittance of the system represented by the following signal flow graph :

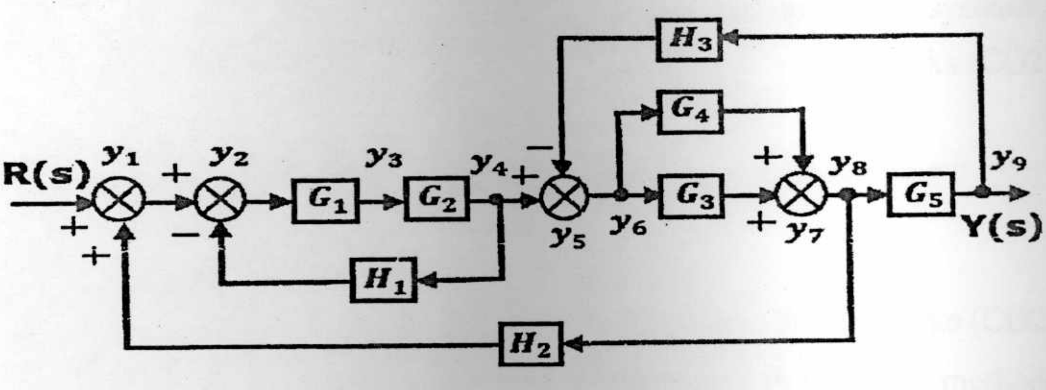


where, y_1 is the input variable and y_4 is the output variable.

10 Marks (CO3)

OR

(b) Construct the signal flow graph of the system as represented by the following block diagram : 10 Marks (CO3)



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